MAJOR PROJECT ARCHITECTURE | STATUS: VERIFIED & APPROVED | TARGET: MEDICAL NECESSITY & WAITING PERIOD

End-to-End Dataset & Evaluation Implementation Plan

Master Guide: Source Datasets, Golden Benchmarks, and 3-Tier DeepEval Evaluation Suite

Executive Summary & Status

1. Downloaded Policy Wordings Status (`data/policy_wordings/`)

All **13 essential retail policy documents across India's top 4 private health insurers (ICICI Lombard, Care Health, HDFC ERGO, Star Health)** are organized and verified in `data/policy_wordings/` :

#	Insurer	File Name	Size	Product Category	Relevance to Pipelines
1	ICICI Lombard	`complete-health-insurance.pdf`	416 KB	Retail Comprehensive	**Primary Core**: Standard exclusions, 24/48-month PED clauses
2	ICICI Lombard	`health-shield-360-retail...pdf`	1.3 MB	Retail Modular	**Primary Core**: Elective surgery & exception wording
3	ICICI Lombard	`arogya-sanjeevani.pdf`	549 KB	IRDAI Standardized	**Primary Core**: Statutory benchmark (Code-Excl01 to Excl04)
4	ICICI Lombard	`elevate.pdf`	1.6 MB	Modern Comprehensive	**Primary Core**: Modern treatments, robotic surgeries, infinite SI
5	ICICI Lombard	`health-advantedge...pdf`	1.3 MB	High-Sum-Insured Retail	High-value claim sub-limits and room-rent clauses
6	ICICI Lombard	`health-booster...pdf`	626 KB	Super Top-Up	Deductible threshold calculations and secondary claim denials
7	ICICI Lombard	`max-protect.pdf`	899 KB	Retail Health	1Y/2Y/3Y waiting period modification rules
8	Care Health	`care...pdf`	2.1 MB	Classic Comprehensive	Flagship retail comprehensive coverage (formerly Religare Care)
9	Care Health	`care-supreme...pdf`	2.2 MB	Modern Comprehensive	Modern daycare, robotic surgery, cumulative bonus rules
10	HDFC ERGO	`hdfc-optima-secure.pdf`	925 KB	Modern Flagship	Automatic 4X restore benefit, non-medical expense coverage
11	HDFC ERGO	`hdfc-optima-restore.pdf`	773 KB	Classic Flagship	Most widely held retail health policy in HDFC ERGO history
12	Star Health	`star-comprehensive.pdf`	2.1 MB	Standalone Flagship	#1 retail health insurer in India, bariatric & modern surgery
13	Star Health	`star-family-health-optima.pdf`	1.3 MB	Classic Floater	Widely held family floater health policy in India

> **NOTE**
> All irrelevant policy types (Group policies, Personal Accident, Hospital Cash, Single-disease benefit riders) have been excluded, ensuring our knowledge base and evaluation suite remain uncontaminated.

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2. Complete Source Dataset Registry (Free & Public)

To build both the **Runtime Knowledge Base** and the **3-Tier Golden Evaluation Datasets**, we utilize 13 public, freely accessible datasets:

Part A: Raw Source Pools & Knowledge-Base Inputs (D1 – D8)

ID	Dataset Name	Source / Provider	License	Role in System & Lifecycle
D1	CMS DE-SynPUF	Kaggle (CMS)	Public Domain	**Upstream Pool Only**: Sampled to synthesize golden claim dates and amounts; *dropped at runtime*.
D2	MTSamples	Kaggle (CC0)	CC0 Public	**Upstream Pool Only**: Sampled to populate clinical narratives in golden JSONs; *dropped at runtime*.
D3	ICD-10-CM Codes	Kaggle / CDC	Public Domain	**Active Runtime CSV** (`icd10_codes.csv`): Diagnosis validation & entity lookup.
D4	ClaimSaathi	Kaggle	Open	**Upstream Pool Only**: Sampled for Indian waiting period edge cases; *dropped at runtime*.
D5	HCPCS Level II	CMS.gov	Public Domain	**Active Runtime CSV** (`procedure_codes.csv`): Procedure code verification.
D6	IBM Policy Rules	GitHub (IBM)	Apache 2.0	Reference schema for converting policy clauses to IF-THEN rules.
D7	13 Retail Policy PDFs (4 Insurers)	Local (`data/policy_wordings/`)	Free Public	**Active Runtime Core**: Extracted into `policy_clauses.csv` (Vector Store for RAG, 1,997 clauses) & `nme_exclusions.csv` .
D8	CARC/RARC Codes	X12 / CMS	Public Domain	**Active Runtime CSV** (`denial_reasons.csv`): Taxonomy mapping table for the Denial Router.

Part B: Dedicated Golden Evaluation Benchmarks (D9 – D13)

ID	Dataset Name	Source / Provider	License	Role in Evaluation
D9	CA DMHC IMR Trend	CA Open Data / Kaggle	Public Domain	**100,000+ real Medical Necessity appeal cases** with clinical findings and Overturned/Upheld ground truth. Direct CSV download.
D10	Persius/imr-appeals & hicric	Hugging Face	Open Research	Span-labeled external appeal adjudications and policy rules for health coverage denials. Direct download.
D11	CIO Ombudsman & Indian Kanoon	cioins.co.in / Indian Kanoon	Public Legal Records	**Real Indian Pre-Existing Disease (PED) & waiting period repudiations**. Annual Reports direct PDF download; Indian Kanoon full legal texts (zero OTP/login).
D12	InsLogicBench & InsClaimBench	ACL 2026 / arXiv:2610.09671	Open Benchmark	Nested Toulmin argumentation ground truth & claim decision chains.
D13	IRDAI Indian Repudiation Notices	IRDAI Guidelines / Insurer Portals	Public Regulatory	Standard 1-page health claim repudiation notice format (ICICI Lombard, Star Health) populated with ClaimSaathi & Kanoon facts. *(100% pure healthcare; replaces generic Nemotron-PII)*.

Part C: Accessibility & Verification Audit Table

Every source dataset has been verified for accessibility, licensing, and domain relevance:

ID	Dataset	Direct Download Link / Access	Cost	Auth / Credentials Needed?	Domain Pure Healthcare?	Verified Status
D1	CMS DE-SynPUF	Kaggle	Free	Free Kaggle Account	Yes (Medicare Claims)	**VERIFIED**
D2	MTSamples	Kaggle	Free	Free Kaggle Account	Yes (Clinical Notes)	**VERIFIED**
D3	ICD-10-CM Codes	CDC / CMS	Free	None (Public Domain)	Yes (Diagnosis Taxonomy)	**VERIFIED**
D4	ClaimSaathi	Kaggle	Free	Free Kaggle Account	Yes (Indian Health Claims)	**VERIFIED**
D5	HCPCS Level II	CMS.gov	Free	None (Direct Excel/CSV)	Yes (Medical Procedures)	**VERIFIED**
D6	IBM Policy Rules	GitHub	Free	None (Public GitHub)	Yes (Health Policy Rules)	**VERIFIED**
D7	13 Policy PDFs (4 Insurers)	Local: `data/policy_wordings/`	Free	None (Downloaded & Verified)	Yes (Retail Health Policies)	**READY LOCALLY**
D8	CARC/RARC Codes	X12.org / CMS	Free	None (Public Domain)	Yes (Denial Reasons)	**VERIFIED**
D9	CA DMHC IMR Trend	CA Open Data Portal	Free	None (Direct CSV Download)	Yes (Medical Necessity)	**VERIFIED**
D10	Persius/imr-appeals	Hugging Face	Free	None (Direct HF Download)	Yes (Health Coverage Appeals)	**VERIFIED**
D11	CIO & Indian Kanoon	Indian Kanoon & CIO Publications	Free	None (Zero OTP / Open Web)	Yes (Indian PED Repudiations)	**VERIFIED**
D12	InsLogicBench	ACL Anthology / arXiv:2610.09671	Free	None (Open Research)	Yes (Insurance Reasoning)	**VERIFIED**
D13	IRDAI Repudiation Forms	IRDAI Portals	Free	None (Regulatory Templates)	Yes (Health Claim Repudiation)	**VERIFIED**

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3. Runtime Knowledge Base vs. Golden JSON Architecture

A critical architectural distinction is made between **Runtime Knowledge Bases** and **Evaluation Benchmarks**:

- **Runtime Knowledge-Base CSVs:** External dictionaries and vector databases that AI agents query during live execution (e.g., searching for clause wording, checking ICD-10 validity).
- **Golden Dataset JSONs:** Self-contained evaluation cases containing both the input prompt/documents and the expected ground truth used by **DeepEval** to grade the system.

<!-- DIAGRAM_1 -->

A. The 5 Active Runtime Knowledge-Base CSVs

Only these 5 CSVs are maintained at runtime. Agents query these files as external knowledge:

```
c:\Users\Saketh\Desktop\Major Project\data\runtime\
├── policy_clauses.csv       (Extracted from D7 PDFs: 1,997 structured clauses indexed in Vector DB across 4 insurers)
├── denial_reasons.csv       (Standardized D8 + IRDAI: ~80 denial categories for Router lookup)
├── icd10_codes.csv          (Direct D3: 70,000+ diagnosis codes for medical entity validation)
├── procedure_codes.csv      (Direct D5: 7,000+ procedure codes for surgery validation)
└── nme_exclusions.csv       (Extracted from D7 Annexures: ~120 non-medical items for financial audit)
```

Detailed Schemas for Active Runtime CSVs:

1. `policy_clauses.csv` (Loaded into ChromaDB / FAISS for RAG):
   - Columns: `clause_id`, `insurer`, `product_name`, `section`, `clause_number`, `clause_text`, `clause_type`, `has_exception`, `exception_text`, `temporal_rule`
2. `denial_reasons.csv` (Used by Denial Reason Normalization & Router):
   - Columns: `reason_id`, `category`, `sub_category`, `reason_text`, `carc_code`, `typical_clause_type`, `appeal_route`
3. `icd10_codes.csv` (Used by Medical Entity Linker):
   - Columns: `code`, `short_description`, `long_description`, `is_billable`

4. `procedure_codes.csv` (Used by Procedure Verification):
   - Columns: `hcpcs_code`, `short_description`, `category`
5. `nme_exclusions.csv` (Used by Financial Audit Module):
   - Columns: `item_id`, `item_name`, `category`, `typically_excluded`

B. Eliminated / Redundant CSVs (Superseded by Golden JSONs)

The following 2 CSVs from initial planning are **completely eliminated** from runtime:

Eliminated File	Original Purpose	Why It Is Eliminated
❌ `medical_notes.csv`	Raw table of 2,000 MTSamples clinical notes.	**Superseded by Golden JSONs.** Clinical discharge summaries are embedded directly inside `golden_clinical_evidence.json` and `golden_pipeline_medical_necessity.json`. Agents receive the note directly in the test case; querying a 2,000-row CSV at runtime is unnecessary.
❌ `claims_master.csv`	10,000 synthetic patient rows.	**Superseded by Golden JSONs.** Patient demographics, policy inception dates, waiting period milestones, and billed amounts are self-contained within `golden_chronology_ped.json` and `golden_intake_entities.json`.

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4. The 3-Tier Golden Datasets Specification

To evaluate our two prototype pipelines (**Medical Necessity** and **Pre-Existing Waiting Period**) and the overall system, we construct **10 purpose-built Golden Datasets**.

<!-- DIAGRAM_2 -->

Tier 1: Component-Wise Golden Datasets

G1: `golden_intake_entities.json` (Component 1.1: Denial Intake)

- **Target Component:** Denial Intake & Entity Extraction Agent ([Lop21])
- **Source:** D10 (Persius/imr-appeals for Medical Necessity) + D13 (IRDAI Repudiation Notices populated with D4 ClaimSaathi & Kanoon facts for Waiting Period).
- **Size:** 100 cases (unstructured text letters).
- **Schema:**


```
{
  "case_id": "G1-001",
  "raw_letter_text": "Ref: CLM-2024-8921. Dear Ramesh Kumar, with reference to your hospitalization on 15/03/2024 at Apollo Hos
  "ground_truth_entities": {
    "claim_id": "CLM-2024-8921",
    "patient_name": "Ramesh Kumar",
    "denial_date": "2024-03-15",
    "procedure": "Total Knee Replacement",
    "icd10_code": "M17.11",
    "claimed_amount": 450000.0,
    "cited_clause_raw": "Clause 4.3",
    "denial_category_hint": "Elective Surgery"
  }
}
```


G2: `golden_clause_retrieval.json` (Component 1.2: Policy Clause RAG)

- **Target Component:** Policy Clause Retriever & Verifier ([Pic22], [Pok26])
- **Source:** Extracted directly from the 13 verified retail policy PDFs across ICICI Lombard, Care Health, HDFC ERGO, and Star Health.
- **Size:** 100 query-context triplets.
- **Schema:**

```
{
  "query_id": "G2-018",
  "user_query": "Is Total Knee Replacement covered if conservative medical treatment failed under Health Shield 360?",
  "product_name": "Health Shield 360",
  "expected_clause_id": "HS360-EX-043",
  "expected_subclause": "4.3.2",
  "has_exception": true,
  "ground_truth_context_snippet": "Sub-clause 4.3.2: Elective procedures shall not include surgeries where conservative medical
  "distractor_clauses": ["HS360-EX-001", "HS360-COV-012"]
}
```


G3: `golden_clinical_evidence.json` **(Component 1.3A: Clinical Extraction)**

- **Target Component:** Clinical Evidence Investigation Agent ([Pan24], [Ahs23])
- **Source:** D9 (California DMHC IMR dataset) + D2 (MTSamples discharge summaries).
- **Size:** 60 annotated clinical narratives.
- **Schema:**


```
{
  "case_id": "G3-005",
  "clinical_notes": "58yo male with chronic right knee pain unresponsive to 18 months of physical therapy, multiple steroid inj
  "ground_truth_checklist": {
    "conservative_treatment_attempted": true,
    "conservative_treatment_duration_months": 18,
    "treatment_modalities": ["physical therapy", "steroid injections", "NSAIDs"],
    "disease_severity_grade": "Grade IV / Bone-on-bone",
    "physician_necessity_stated": true
  },
  "verbatim_evidence_spans": [
    "unresponsive to 18 months of physical therapy, multiple steroid injections, and daily NSAIDs",
    "Grade IV osteoarthritis with complete joint space loss and bone-on-bone contact"
  ]
}
```


G4: `golden_chronology_ped.json` **(Component 1.3B: Chronology Engine)**

- **Target Component:** Chronology & Waiting Period Engine ([Pok26])
- **Source:** D11 (Indian Insurance Ombudsman awards) + D4 (ClaimSaathi timeline records).
- **Size:** 60 chronological disputes.
- **Schema:**


```
{
  "case_id": "G4-012",
  "policy_inception_date": "2021-04-10",
  "admission_date": "2024-05-15",
  "claimed_condition": "Hypertensive Nephropathy",
  "clinical_history_onset_statement": "Patient has a history of essential hypertension diagnosed 2 years ago (May 2022).",
  "proposal_form_ped_declared": false,
  "continuous_renewal_years": 3,
  "applicable_waiting_period_months": 36,
  "ground_truth_calculation": {
    "elapsed_months": 37.1,
    "first_symptom_relative_to_inception": "POST_INCEPTION",
    "waiting_period_satisfied": true,
    "pre_existing_non_disclosure": false,
    "moratorium_clause_applicable": false,
    "symbolic_decision": "DENIAL_REVERSAL_WAITING_PERIOD_EXPIRED"
  }
}
```


G5: `golden_reasoning_toulmin.json` **(Component 1.4: Toulmin Reasoner)**

- **Target Component:** Toulmin & Defeasible Reasoner ([Liu26], [Pok26])
- **Source:** Modeled from D12 (InsLogicBench / InsClaimBench architecture).
- **Size:** 80 structured reasoning cases (50% Reversal, 50% Uphold).
- **Schema:**

```
{
  "case_id": "G5-024",
  "category": "MEDICAL_NECESSITY",
  "toulmin_components": {
    "claim": "Total Knee Replacement is an excluded elective procedure.",
    "grounds": "Patient underwent 18 months of failed conservative management.",
    "warrant": "Clause 4.3 excludes elective surgeries unless conservative treatment failed for >= 6 months.",
    "exception_status": "SATISFIED",
    "rebuttal": "No contradicting evidence found; physician certifies necessity."
  },
  "expected_verdict": "CONTRADICTS_DENIAL",
  "confidence_threshold": 0.90
}
```


G6: `golden_appeal_drafts.json` (Component 1.5: Appeal Drafting Agent)

- **Target Component:** Appeal Agent ([Kis24])
- **Source:** Expert appeal letters adapted from D11 (Ombudsman filings) and D10 (Persius).
- **Size:** 40 complete case-to-letter pairs.
- **Schema:**


```
{
  "case_id": "G6-003",
  "case_facts": { "claim_id": "CLM-2024-8921", "clause": "4.3", "exception": "4.3.2", "patient": "Ramesh Kumar" },
  "mandatory_citations": [
    "Clause 4.3.2 of Health Shield 360",
    "Discharge Summary dated 15/03/2024",
    "18 months of documented conservative management"
  ],
  "target_forum": "Insurance Ombudsman",
  "reference_appeal_text": "To: The Insurance Ombudsman, Hyderabad...\nSubject: Appeal against wrongful repudiation of Claim CL
}
```

Tier 2: Pipeline-Wise Golden Datasets (End-to-End Category Benchmarks)

G7: `golden_pipeline_medical_necessity.json` (Pipeline A Benchmark)

- **Target:** End-to-End Medical Necessity Pipeline (Intake $\rightarrow$ RAG $\rightarrow$ Clinical Extraction $\rightarrow$ Toulmin $\rightarrow$ Appeal).
- **Source:** 50 real clinical dispute cases from **D9 (California DMHC IMR Trend Dataset)** paired with ICICI policy wordings.
- **Composition:**
  - 35 Overturn cases (clear evidence of conservative failure, acute severity, or clinical guideline conformance).
  - 15 Upheld cases (cosmetic surgeries, zero prior conservative therapy, unproven experimental procedures).
- **Evaluation Target:** Complete pipeline must take raw unstructured inputs and output:
  1. Correct binary verdict (`OVERTURN` vs `UPHOLD`).
  2. Grounded Evidence Matrix with >0.85 recall on critical clinical facts.
  3. Grounded appeal draft.

G8: `golden_pipeline_waiting_period.json` (Pipeline B Benchmark)

- **Target:** End-to-End Pre-Existing Waiting Period Pipeline (Intake $\rightarrow$ RAG $\rightarrow$ Chronology Engine $\rightarrow$ Toulmin $\rightarrow$ Appeal).
- **Source:** 50 real cases curated from **D11 (Indian Insurance Ombudsman Awards & Indian Kanoon)**.
- **Composition:**
  - 25 Reversals where waiting period expired prior to admission.
  - 10 Reversals protected by the statutory Moratorium Clause (IRDAI 8-year / 60-month rule).
  - 15 Justified Denials (deliberate non-disclosure of bypass surgery 2 months before policy inception).
- **Evaluation Target:** Symbolic math accuracy $= 100%$, statutory compliance $= 100%$.

Tier 3: Application-Wise Golden Datasets

G9: `golden_denial_router.json` (7-Class Triage Router)

- **Target:** Global Denial Router (Triage Node 4 in architecture).
- **Source:** D8 (CARC/RARC codes) + IRDAI claim rejection taxonomy.
- **Size:** 140 raw denial letter paragraphs (20 per category across all 7 categories: `MEDICAL_NECESSITY`, `PRE_EXISTING_WAITING_PERIOD`, `POLICY_EXCLUSION`, `FINANCIAL_LIMIT`, `DOCUMENTATION`, `ADMINISTRATIVE_PROCEDURAL`, `ELIGIBILITY_COVERAGE`).
- **Ground Truth:** Single-label categorical enum.

G10: `golden_hitl_benchmark.json` (Human Adjudicator Review Alignment)

- **Target:** System-wide Human Review Node (Node 14 in architecture, [Sbo24]).
- **Size:** 30 comprehensive claim appeal packs.
- **Ground Truth:** Independent adjudication consensus by two senior medical claim reviewers scoring Appeal Persuasiveness, Factual Correctness, and Reversal Feasibility.

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5. Golden Dataset Construction Pipeline (Step-by-Step)

Here is the exact step-by-step workflow to build these datasets from raw downloads:

<!-- DIAGRAM_3 -->

Scripted Construction Details:

1. **Curate Policy Clauses (`policy_clauses.csv` & G2):**
   - Run Python PyMuPDF (`fitz`) over `complete-health-insurance.pdf`, `health-shield-360-retail`, `arogya-sanjeevani`, and `elevate`.
   - Parse Section 3 (Waiting Periods) and Section 4 (Exclusions). Extract rules and sub-clauses into structured JSON/CSV.
2. **Curate Clinical Evidence (G3 & G7):**
   - Download `Independent_Medical_Review_IMR_Determinations_Trend.csv` from CA Open Data portal (free, ~150 MB).
   - Filter `Reason_For_Dispute == "Medical Necessity"` and `Diagnosis_Category.isin(["Orthopedic", "Cardiovascular", "Neurology"])`.
   - Take 50 cases with rich `Findings` text. The `Determination_Outcome` column (`Overturned` vs `Upheld`) gives verified ground truth.
3. **Curate Chronology Benchmarks (G4 & G8):**
   - Extract 50 case summaries from Indian Kanoon querying `"Insurance Ombudsman" AND "waiting period"`.
   - Record exact dates, waiting periods cited, and the Ombudsman's binding award.
4. **Curate Router Benchmark (G9):**
   - Map 140 CARC/RARC code descriptions and real rejection notices to the 7 target categories.

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6. Three-Tier Evaluation Framework & Metrics

We evaluate performance using standard statistical metrics, symbolic exact-matches, and **DeepEval** metrics:

Level	Component	Evaluation Metric	Tool / Formula	Target Threshold
L1	Intake Entities (G1)	Token F1 & Exact Match	`scikit-learn` / Python	Entity F1 ≥ 0.95, EM ≥ 0.98
L1	Clause RAG (G2)	`ContextualPrecisionMetric` `ContextualRecallMetric`	**DeepEval**	Precision ≥ 0.90 Recall ≥ 0.95
L1	Clinical Extraction (G3)	`FaithfulnessMetric` Checklist Accuracy	**DeepEval** $\frac{\text{Correct Checks}}{\text{Total Checks}}$	Faithfulness ≥ 0.95 Checklist Acc ≥ 0.90
L1	Chronology Engine (G4)	Deterministic Exact Match Date Offset Error (MAE)	Python Unit Tests $\text{MAE} = \frac{1}{N}\sum \|d_{\text{pred}} - d_{\text{true}}\|$	Accuracy $= 100\%$ $\text{MAE} = 0$ days
L1	Toulmin Reasoner (G5)	Verdict Accuracy `GEval` (Logic Soundness)	`scikit-learn` **DeepEval** G-Eval (1-5)	Accuracy ≥ 0.92 G-Eval $\ge 4.5/5.0$
L1	Appeal Drafter (G6)	`FaithfulnessMetric` `GEval` (Legal Persuasiveness)	**DeepEval**	Faithfulness ≥ 0.98 Persuasiveness $\ge 4.2/5$
L2	Medical Necessity (G7)	Overturn Prediction F1 Clinical Coverage Recall	End-to-End Suite	Macro-F1 ≥ 0.90 Coverage Recall ≥ 0.85
L2	Waiting Period (G8)	Statutory Compliance Rate Ombudsman Alignment	End-to-End Suite	Compliance $= 100\%$ Alignment ≥ 0.92
L3	Denial Router (G9)	Multiclass Macro-F1 Confusion Matrix	`scikit-learn`	Macro-F1 ≥ 0.95
L3	HITL Review (G10)	Inter-Rater Reliability	**Cohen's Kappa (κ)** $\kappa = \frac{P_o - P_e}{1 - P_e}$	$\kappa \ge 0.80$ (Substantial Agreement)
L3	Operational	Latency & Token Cost	Profiler & Token Counter	P95 $< 45\text{s}$, Cost $< 0.10

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7. DeepEval Code Harness & Execution Blueprint

Below is the concrete implementation harness using `deepeval` to run component and pipeline evaluations:

```python
import json
from deepeval import evaluate
from deepeval.metrics import (
    ContextualPrecisionMetric,
    ContextualRecallMetric,
    FaithfulnessMetric,
    GEval,
)
from deepeval.test_case import LLMTestCase, LLMTestCaseParams

# ------------------------------------------------------------
# 1. Custom G-Eval: Toulmin Logical Argumentation Soundness
# ------------------------------------------------------------
toulmin_argument_metric = GEval(
    name="Toulmin Argumentation Soundness",
    criteria=(
        "Evaluate whether the reasoning correctly establishes that an exception defeats the policy exclusion. "
        "The conclusion must logically flow from the grounds and warrant without contradictions or ungrounded assumptions."
    ),
    evaluation_params=[
        LLMTestCaseParams.INPUT,
        LLMTestCaseParams.ACTUAL_OUTPUT,
        LLMTestCaseParams.RETRIEVAL_CONTEXT,
    ],
    threshold=0.85,
)

# ------------------------------------------------------------
# 2. Custom G-Eval: Ombudsman Appeal Persuasiveness & Regulatory Rigor
# ------------------------------------------------------------
appeal_quality_metric = GEval(
    name="Ombudsman Appeal Legal Rigor",
    criteria=(
        "Assess whether the draft appeal: (1) cites the exact policy sub-clause and exception, "
        "(2) references specific clinical reports and treatment duration, and (3) adheres to formal Insurance Ombudsman standards.
    ),
    evaluation_params=[
        LLMTestCaseParams.INPUT,
        LLMTestCaseParams.ACTUAL_OUTPUT,
        LLMTestCaseParams.EXPECTED_OUTPUT,
    ],
    threshold=0.80,
)

# ------------------------------------------------------------
# 3. Execution Pipeline for Medical Necessity (G7 Benchmark)
# ------------------------------------------------------------
def run_medical_necessity_evaluation(golden_dataset_path: str):
    with open(golden_dataset_path, "r") as f:
        test_data = json.load(f)

    test_cases = []
    faithfulness_metric = FaithfulnessMetric(threshold=0.95)
    context_recall_metric = ContextualRecallMetric(threshold=0.90)

    for item in test_data:
        # Run agent pipeline to get actual outputs
        # actual_output, retrieved_context = medical_necessity_pipeline.run(item["input"])

        tc = LLMTestCase(
            input=item["raw_input_narrative"],
            actual_output=item["pipeline_actual_output"],
            expected_output=item["ground_truth_verdict_and_summary"],
            retrieval_context=item["retrieved_policy_and_clinical_context"],
            context=item["case_facts"],
        )
        test_cases.append(tc)

    # Execute DeepEval test suite
    results = evaluate(
        test_cases=test_cases,
        metrics=[
            faithfulness_metric,
            context_recall_metric,
```

```
            toulmin_argument_metric,
            appeal_quality_metric,
        ],
    )
    return results

if __name__ == "__main__":
    print("Ready to execute 3-tier DeepEval test suite.")
```

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8. Master Traceability Matrix

This table traces every stage from source data to the final evaluation metric:

Flow Stage / Agent	Paper Method	Source Data Used	Golden Dataset Produced	Evaluation Tool & Metric
Intake Agent	[Lop21]	D13 + D8 + D4	`golden_intake_entities.json`	Python Exact Match & Entity F1
Denial Router	[Owo24]	D8 + IRDAI Tax.	`golden_denial_router.json`	Multiclass Confusion Matrix & Macro-F1
Clause Verifier	[Pic22], [Pok26]	D7 (Local PDFs)	`golden_clause_retrieval.json`	DeepEval Contextual Precision & Recall
Clinical Investigator	[Pan24], [Ahs23]	D9 (CA IMR) + D2	`golden_clinical_evidence.json`	DeepEval Faithfulness & Checklist Acc
Chronology Engine	[Pok26]	D11 (Ombudsman) + D4	`golden_chronology_ped.json`	Symbolic Math Acc ($=100%$) & MAE
Toulmin Reasoner	[Liu26]	D12 (InsLogicBench)	`golden_reasoning_toulmin.json`	DeepEval G-Eval Argumentation Soundness
Appeal Drafter	[Kis24]	D11 + D10	`golden_appeal_drafts.json`	DeepEval Faithfulness & Legal Persuasiveness
Med. Nec. Pipeline	End-to-End	D9 + D7 + D2	`golden_pipeline_medical_necessity.json`	Pipeline Overturn Accuracy & Coverage
Waiting Period Pipeline	End-to-End	D11 + D7 + D4	`golden_pipeline_waiting_period.json`	Statutory Compliance & Chronology F1
Human Review	[Sbo24]	All Artifacts	`golden_hitl_benchmark.json`	Cohen's Kappa ($\kappa \ge 0.80$)